University of Bremen

Deutsches Zentrum für Luft- und Raumfahrt
German Aerospace Center

Smart Sensor Fusion for Robust Initialization and Accurate Trajectory Estimation

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Overview

Where we are

Motivation & Problem Statement

Why fuse IMU $\times$ GNSS?

Single-sensor limits

- GNSS: dropouts, multipath, low rate (1–10 Hz), latency.
- IMU: bias-driven drift (gyro $\rightarrow$ attitude; accel $\rightarrow$ position $\propto t^2$).

Complementary strengths

- IMU: high-rate, smooth propagation through GNSS gaps.
- GNSS: absolute updates to bound drift and observe biases.

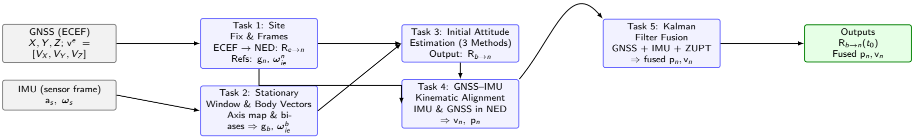
Problem statement

- Initialize the navigation frame correctly and estimate initial attitude.
- Put GNSS and IMU in the *same* frame (NED) with consistent kinematics.
- Fuse to obtain drift-bounded $\mathrm{p}_n, \mathrm{v}_n$ with reliable covariances.

Key insight

Robust initial alignment *and* continuous fusion are mandatory for stable, accurate trajectory estimation.

End-to-End Pipeline: Tasks 1–5



Legend: Gray = inputs; Blue = processing (Tasks 1–5); Green = outputs.

Where we are

Site Fix & Frames from GNSS (Task 1, Subtask 1.1)

Inputs: GNSS X, Y, Z (ECEF) or (ϕ, λ, h) (geodetic), WGS-84.

WGS-84 constants:

$$a = 6378137 \text{ m}, \quad f = \frac{1}{298.257223563}, \quad e^2 = f(2 - f)$$

ECEF → Geodetic:

$$\lambda = \operatorname{atan2}(Y, X), \quad p = \sqrt{X^2 + Y^2}$$

$$N(\phi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi}}, \quad h = \frac{p}{\cos \phi} - N(\phi)$$

ECEF → NED rotation:

$$\mathrm{C}_n^e = \begin{bmatrix} -\sin\phi\cos\lambda & -\sin\phi\sin\lambda & \cos\phi \\ -\sin\lambda & \cos\lambda & 0 \\ -\cos\phi\cos\lambda & -\cos\phi\sin\lambda & -\sin\phi \end{bmatrix}, \quad \mathrm{v}_n = \mathrm{C}_n^e \mathrm{v}_e$$

Outputs: site (ϕ, λ, h); frame map C_n^e.

Task 1: Launch Site

Task 1 — Initial GNSS location

Gravity Model → g_{NED} (Task 1, Subtask 1.2)

Somigliana normal gravity (sea level):

$$\gamma(\phi) = \gamma_e \frac{1 + k \sin^2 \phi}{\sqrt{1 - e^2 \sin^2 \phi}}$$

$$\gamma_e = 9.7803253359 \ \mathrm{m/s^2}, \quad k = 0.00193185265241$$

Free-air correction (height h):

$$\gamma(\phi, h) \approx \gamma(\phi) - 3.086 \times 10^{-6} \, h$$

Reference gravity in NED:

$$\boxed{\mathrm{g}_n = \begin{bmatrix} 0 \\ 0 \\ \gamma(\phi, h) \end{bmatrix}}$$

Measured (body) counterpart from IMU:

$$\mathrm{g}_b = \mathrm{mean}(\text{accelerometers}), \quad \text{norm} \approx \gamma$$

Earth Rotation $\rightarrow$ $\boldsymbol{\omega}_{ie}^{\text{NED}}$ (Task 1, Subtask 1.3)

Earth spin (ECEF):

$$\boldsymbol{\omega}_{ie}^{e} = \begin{bmatrix} 0 \\ 0 \\ \omega_e \end{bmatrix}, \quad \omega_e = 7.292115 \times 10^{-5}\ \mathrm{rad/s}$$

Map to NED:

$$\boldsymbol{\omega}_{ie}^{n} = \omega_e \begin{bmatrix} \cos\phi \\ 0 \\ -\sin\phi \end{bmatrix}$$

Measured (body) counterpart:

$$\boldsymbol{\omega}_{ie}^{b} = \mathrm{mean}(\text{gyros}), \quad \text{norm} \approx \omega_e$$

Outputs: $\boldsymbol{\omega}_{ie}^{n}$ for Wahba; $\boldsymbol{\omega}_{ie}^{b}$ from IMU.

Reference Validation (Physics & Geometry) (Task 1, Subtask 1.4)

Magnitude Checks:

$$\|\mathrm{g}_n\| \overset{?}{\approx} \gamma(\phi, h), \qquad \|\boldsymbol{\omega}_{ie}^n\| \overset{?}{=} \omega_e$$

Hemisphere sign (Southern):

$$\phi < 0 \;\Rightarrow\; \omega_D = (-\omega_e \sin\phi) > 0$$

Rotation matrix sanity:

$$\mathrm{C}_n^e (\mathrm{C}_n^e)^\top = \mathrm{I}, \quad \det(\mathrm{C}_n^e) = +1$$

Static window checks (IMU):

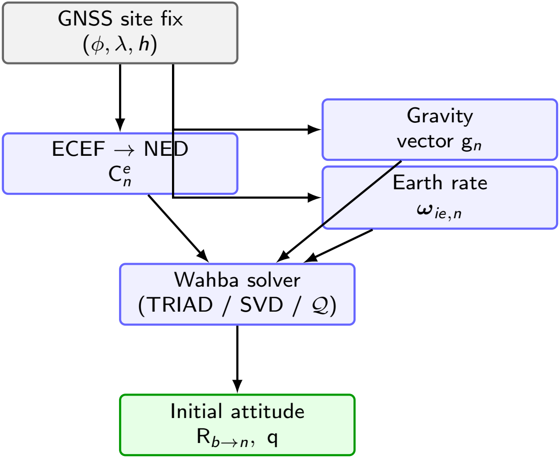
$$\|\mathrm{g}_b\| \approx \gamma, \quad \|\boldsymbol{\omega}_{ie}^b\| \approx \omega_e$$

Angle consistency (diagnostic):

$$\theta(\mathrm{a}, \mathrm{b}) = \arccos\left(\frac{\mathrm{a} \cdot \mathrm{b}}{\|\mathrm{a}\|\,\|\mathrm{b}\|}\right)$$

Outputs: validated references & stationary stats

Task 1: Box Workflow Summary



Legend: Gray = inputs Blue = processing Green = outputs

Where we are

Task 1 $\rightarrow$ Task 2 Bridge: What Moves Forward

- Task 1 fixes the navigation frame (NED) at the launch site using GNSS.
- Reference vectors computed in NED: gravity g_n, Earth rate $\boldsymbol{\omega}_{ie,n}$.
- These vetted references are the "world" vectors for Wahba (Task 3).
- Task 2: measure their **body-frame** counterparts in a stationary window.

GNSS Site: $\varphi = -31.871173^\circ,\ \lambda = 133.455811^\circ$
Reference Vectors:

$$g_n = [0,\ 0,\ 9.7942]\ \mathrm{m/s^2}, \quad \boldsymbol{\omega}_{ie,n} = [6.19 \times 10^{-5},\ 0,\ 3.85 \times 10^{-5}]\ \mathrm{rad/s}$$

- Detect a **stationary interval** with minimal variance (acc/gyro).
- Use this window to compute stable body-frame means:

$$\mathrm{g}_b, \quad \boldsymbol{\omega}_{ie,b}$$

- Longer windows $\rightarrow$ lower noise; must exclude all motion.

Selection Stats:

- Index range: `400 : 479907` (N = 479,508 samples)
- Duration: **1198.77 s** of **1250.00 s** ($\approx$ **95.9%**)
- Sanity: $\mathrm{var}(\mathrm{a})$ low, $\mathrm{var}(\boldsymbol{\omega})$ low, $\|\mathrm{a}\| \approx |g|$

Task 2.2: Sensor-to-Body Axis Mapping

- Map raw sensor axes into vehicle body axes via matrix:

$$C_{bs} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}, \qquad v_b = C_{bs}\, v_s$$

Component mapping:

- $b_x = -s_y,\ b_y = -s_z,\ b_z = s_x$
- Acceleration $\rightarrow$ m/s^2, Angular rate $\rightarrow$ rad/s

Average mapped values over the static window.

Task 2.2: Axis Plot

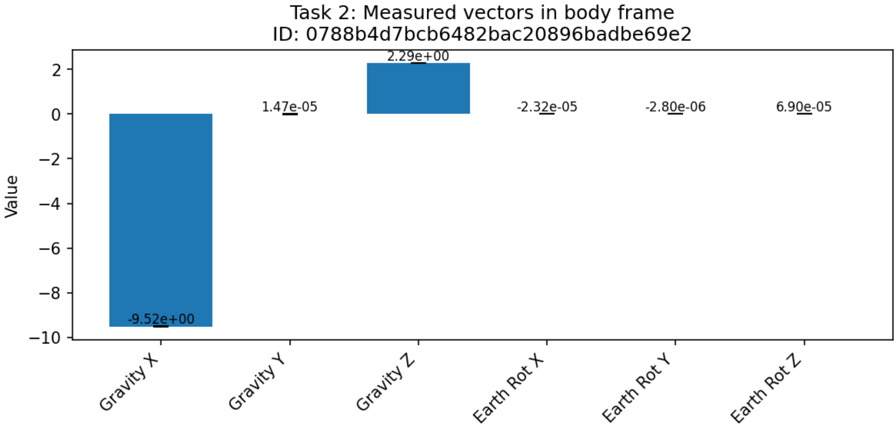


Figure: Vectors

Task 2.3: Body-Frame Vector Estimates (Averaged)

- Compute static mean values for use in Wahba problem (Task 3):

$$\mathrm{g}_b = \begin{bmatrix} -1.51 \times 10^{-5} \\ -2.2908 \\ -9.5226 \end{bmatrix} \ \mathrm{m/s^2} \qquad \boldsymbol{\omega}_{ie,b} = \begin{bmatrix} 2.80 \times 10^{-6} \\ -6.90 \times 10^{-5} \\ -2.32 \times 10^{-5} \end{bmatrix} \ \mathrm{rad/s}$$

These means become the **measured body-frame vectors** for initial alignment.

Task 2.4: Validation Gates (Pass Before Wahba)

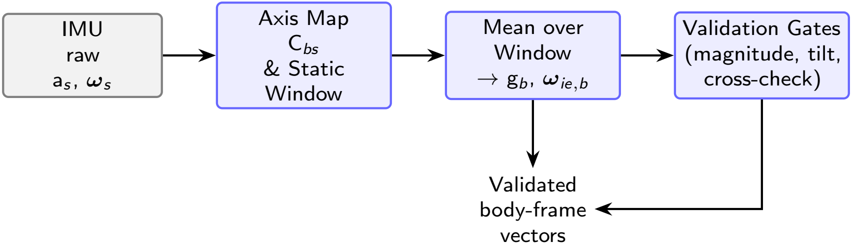
- **Magnitude checks:**
  - $\|g_b\| \approx 9.8 \text{ m/s}^2$
  - $\|\boldsymbol{\omega}_{ie,b}\| \approx 7.292 \times 10^{-5}$ rad/s
- **Orientation sanity:** static tilt from body $+Z$ should be small
- **Cross-check:** mapped mean vs $[0,\ 0,\ +g]_n$

Sample:

- Tilt from $+Z_b$: **13.53**$^\circ$ $\rightarrow$ possible mount issue
- Error to $[0,\ 0,\ +g]^n$: **2.307** (diagnostic, unitless)

Validated vectors passed to Task 3 (Wahba initial alignment).

Task 2: Block Workflow Summary



Legend: Gray = input; Blue = processing; Green = outputs.

Where we are

Task 3.1: Initial Attitude via Wahba's Problem

Goal: Estimate the initial body→NED attitude $R_{b\to n}$ that aligns measured body vectors with known NED references.

Inputs:

- **NED references (Task 1):** $g_n = [0,\ 0,\ 9.7942]$ m/s^2; $\boldsymbol{\omega}_{ie,n} = [6.1927\times 10^{-5},\ 0,\ 3.8503\times 10^{-5}]$ rad/s
- **Body-frame means (Task 2):** $g_b \approx [-1.51\times 10^{-5},\ -2.2908,\ -9.5226]$ m/s^2, $\boldsymbol{\omega}_{ie,b} \approx [2.80\times 10^{-6},\ -6.90\times 10^{-5},\ -2.32\times 10^{-5}]$ rad/s

Wahba's Problem: Find R minimizing

$$J(R) = \frac{1}{2}\sum_i a_i \|v_{i,n} - R v_{i,b}\|^2$$

Methods: TRIAD, Davenport $\mathcal{Q}$-method, SVD.

Outputs: $R_{b\to n}$, quaternion q, alignment error vs truth.

Task 3.2: TRIAD Method (Fast, Deterministic)

Orthonormal triads:

$$\hat{t}_1 = \frac{g}{\|g\|}$$
$$\hat{t}_2 = \frac{\hat{t}_1 \times \boldsymbol{\omega}_{ie}}{\|\hat{t}_1 \times \boldsymbol{\omega}_{ie}\|}$$
$$\hat{t}_3 = \hat{t}_1 \times \hat{t}_2$$

$$R_{b\to n} = [\hat{t}_{1n}\ \hat{t}_{2n}\ \hat{t}_{3n}][\hat{t}_{1b}\ \hat{t}_{2b}\ \hat{t}_{3b}]^\top$$

Result: $q = [0.085141,\ -0.717948,\ 0.686063,\ -0.081359]$
Gravity error: $2.4 \times 10^{-5\circ}$ Earth-rate: 0.239°

Task 3.3: Davenport's $\mathcal{Q}$-Method

Idea: Solve Wahba via quaternion maximization.

$$\mathsf{B} = \sum_i a_i \, \mathsf{v}_{i,n} \mathsf{v}_{i,b}^\top$$

$$\mathsf{K} = \begin{bmatrix} \mathrm{tr}(\mathsf{B}) & \boldsymbol{\Delta}^\top \\ \boldsymbol{\Delta} & \mathsf{B} + \mathsf{B}^\top - \mathrm{tr}(\mathsf{B})\mathsf{I} \end{bmatrix}$$

Where $\boldsymbol{\Delta} = [B_{23} - B_{32},\ B_{31} - B_{13},\ B_{12} - B_{21}]^\top$
Leading eigenvector $\rightarrow$ optimal quaternion q.

Result: Same as TRIAD.
Validation: Matches TRIAD exactly.

Task 3.4: SVD Method (Orthogonal Procrustes)

Idea: SVD of B, enforce det $= +1$ for R.

$$\mathsf{B} = \mathsf{U}\boldsymbol{\Sigma}\mathsf{V}^\top$$

$$\mathsf{R}_{b\to n} = \mathsf{U}\,\mathrm{diag}(1, 1, \det(\mathsf{U}\mathsf{V}^\top))\,\mathsf{V}^\top$$

Result: $\mathsf{q} = [0.084425,\ -0.718033,\ 0.686151,\ -0.080609]$
Gravity error: 0.1197°, Earth-rate: 0.1197°

Task 3.5: Method Comparison & Handoff

Summary:

Method	**Gravity Error (°)**	**Earth-rate Error (°)**
TRIAD	2.4×10^{-5}	0.239
Davenport	2.4×10^{-5}	0.239
SVD	0.1197	0.1197

All methods give nearly identical initial attitudes. SVD is most balanced.
All are suitable for EKF initialization (error $< 1^\circ$).

Task 3.5: Error Comparison

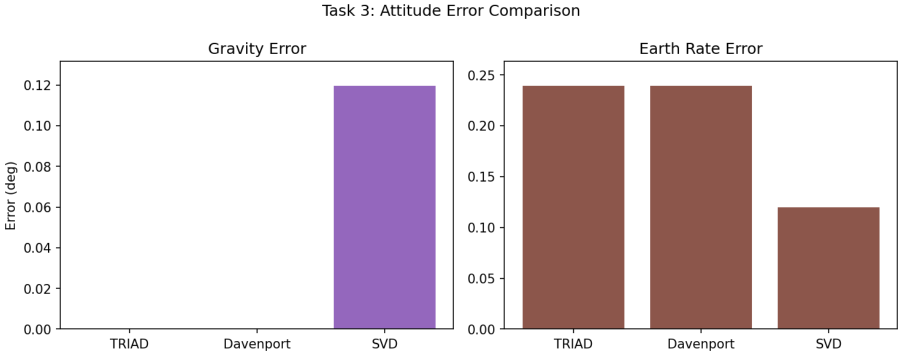
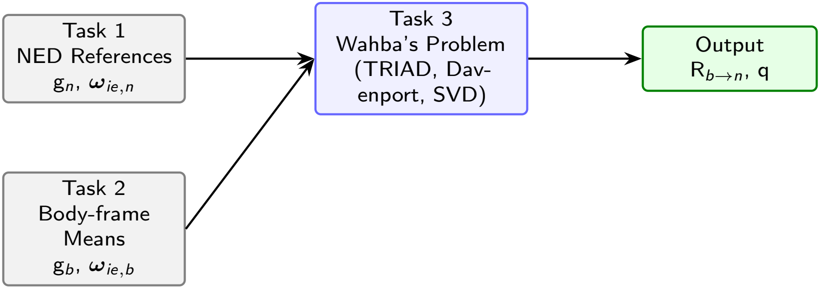


Figure: Altitude Error Comparision

Task 3.6: Workflow Diagram — Initial Attitude Estimation



Legend: Gray = inputs (Task 1,2); Blue = Wahba/Methods; Green = Output for Task 4+5.

Where we are

Task 4.1: GNSS–IMU Data Integration & Comparison

Objective: Using the initial attitude from **Task 3 (Altitude Determination Methods)**, bring GNSS and IMU into the same frame (NED), derive comparable kinematics, and validate consistency before fusion.

Key Steps:

- Use $\mathrm{R}_{b \to n}$ from Task 3
- Transform GNSS ECEF velocity to NED, then estimate GNSS acceleration
- Preprocess IMU data (axis-map, bias correction), rotate to NED, remove gravity
- Integrate IMU acceleration to get $\mathrm{v}_n(t)$, $\mathrm{p}_n(t)$
- Compare GNSS vs IMU in NED, ECEF, Body frames

Key Facts:

- GNSS ECEF vel (1 Hz), p50 = 0.162 m/s
- IMU rate = 400 Hz, static = 95.9%
- Gravity $\mathrm{g}_n = [0, 0, 9.7942]$ m/s^2

Inputs:

- $R_{b \to n}$ (Task 3)
- GNSS reference: $\varphi = -0.556257$ rad, $\lambda = 2.329243$ rad
- Site: $r_0^e = [-3729051.71, 3935675.21, -3348392.35]$ m

ECEF to NED Rotation:

$$R_{e \to n} = \begin{bmatrix} -\sin\varphi\cos\lambda & -\sin\varphi\sin\lambda & \cos\varphi \\ -\sin\lambda & \cos\lambda & 0 \\ -\cos\varphi\cos\lambda & -\cos\varphi\sin\lambda & -\sin\varphi \end{bmatrix}$$

Frames: Body (b), ECEF (e), NED (n)

Task 4.3: GNSS ECEF $\rightarrow$ NED & GNSS Acceleration

GNSS Input:

$$\mathrm{v}^e(t) = [V_X,\ V_Y,\ V_Z]^\top \quad (n = 1250)$$

Transform: $\mathrm{v}^n(t) = \mathrm{R}_{e\rightarrow n}\mathrm{v}^e(t)$

GNSS Acceleration:

$$\mathrm{a}^n(t_k) \approx \frac{\mathrm{v}^n(t_{k+1}) - \mathrm{v}^n(t_{k-1})}{2\,\Delta t}, \quad \Delta t = 1\,\mathrm{s}$$

Task 4.4: IMU Axis Map, Bias, and Gravity Handling

Axis Mapping:

$$C_{s\to b} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}, \quad a_b = C_{s\to b} a_s, \quad \boldsymbol{\omega}_b = C_{s\to b} \boldsymbol{\omega}_s$$

Biases:

$$b_a = [-9.17\times10^{-4},\ 1.99\times10^{-2},\ -4.70\times10^{-3}]\,\mathrm{m/s}^2$$

$$b_g = [-5.51\times10^{-9},\ 1.04\times10^{-7},\ -1.20\times10^{-7}]\,\mathrm{rad/s}$$

Gravity: $g_n = [0,\ 0,\ 9.7942]$ m/s^2; Tilt from $+Z_b$: 13.53°

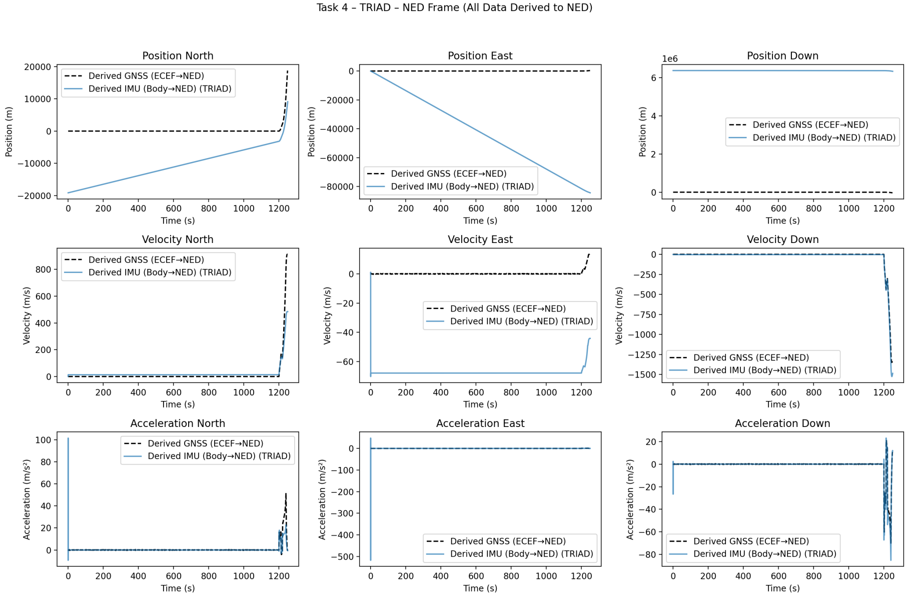
Task 4.5: IMU $\rightarrow$ NED & Kinematic Integration

Steps:

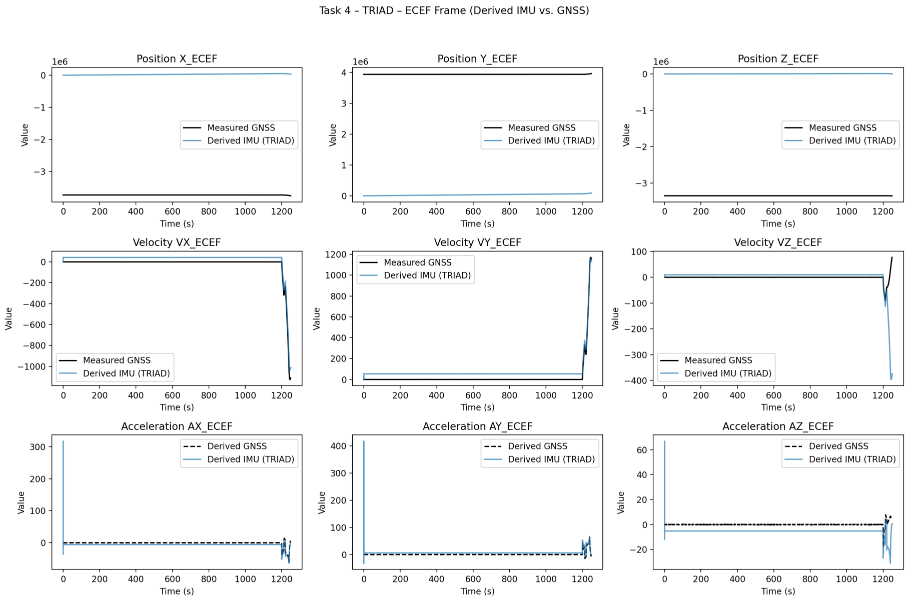
1. De-bias: $\tilde{\mathrm{a}}_b(t) = \mathrm{a}_b(t) - \mathrm{b}_a$
2. Rotate to NED: $\tilde{\mathrm{a}}_n(t) = \mathrm{R}_{b \rightarrow n} \tilde{\mathrm{a}}_b(t)$
3. Remove gravity: $\mathrm{a}^n_{\mathrm{lin}}(t) = \tilde{\mathrm{a}}_n(t) - \mathrm{g}_n$
4. Integrate (trapezoidal):
   $\mathrm{v}^n(t_k) = \mathrm{v}^n(t_{k-1}) + \frac{\Delta t}{2}\left[\mathrm{a}(t_k) + \mathrm{a}(t_{k-1})\right]$
   $\mathrm{p}^n(t_k) = \mathrm{p}^n(t_{k-1}) + \frac{\Delta t}{2}\left[\mathrm{v}(t_k) + \mathrm{v}(t_{k-1})\right]$

IMU rate: 400 Hz ($\Delta t = 0.0025$ s)

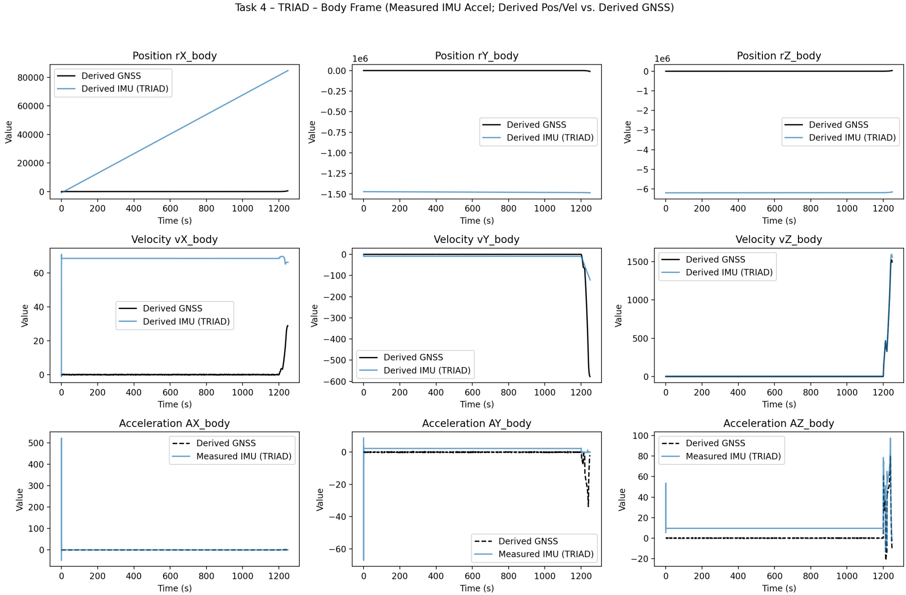
Task 4.6: NED Comparison (GNSS vs IMU)



Task 4.6: ECEF Global Check



Task 4.6: Body-Frame Sanity



Task 4.7: Sanity Checks Before Fusion

Static interval: 400 $\rightarrow$ 479,907 (**95.9%** of 1250 s)

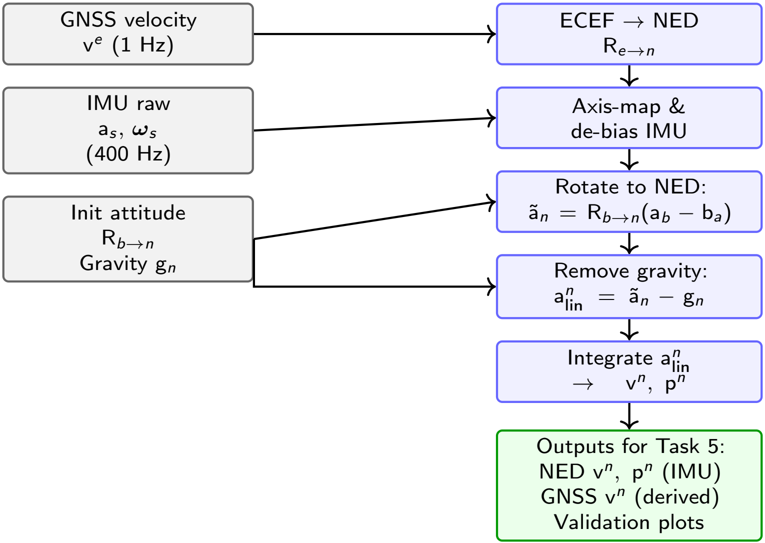
Tilt & residuals:

- Tilt from $+Z_b$: 13.53°
- Residual to $[0, 0, +g]$: 2.3068 (unitless)

Outcome:

- Both sources now in NED
- Alignment sanity confirmed
- Ready for Task 5 (EKF fusion)

Task 4.8: Block Workflow Summary & Handoff



Legend: Gray = inputs Blue = processing Green = outputs

Where we are

Task 5.1: Kalman-Filter Fusion (IMU $\oplus$ GNSS)

Goal: Fuse IMU and GNSS data to estimate drift-bounded $\mathsf{p}_n, \mathsf{v}_n$ with uncertainty.

Inputs:

- $\mathsf{R}_{b\to n}$ from Task 3 (SVD)
- $\mathsf{g}_n = [0,\ 0,\ 9.7942]\ \mathrm{m/s^2}$
- GNSS: ECEF v ($\to$ NED); sampled at 1 Hz
- IMU: $\mathsf{f}_b, \boldsymbol{\omega}_b$, 400 Hz

Key run settings:

- ZUPT thresholds: $a_{\text{thr}} = 0.050$, $g_{\text{thr}} = 0.0050$
- Time shift $= 0.000$ s
- Auto-R: diag pos $= (9.27,\ 9.68,\ 8.13)$, vel $= (25.0,\ 2.46,\ 25.0)$
- Auto-Q: $q_{\text{pos}} = 0.002819$, $q_{\text{vel}} = 1.128$

Task 5.2: State Structure — Position & Velocity Only

State:

$$\mathrm{x}_k = \begin{bmatrix} \mathrm{p}_n \\ \mathrm{v}_n \end{bmatrix} \in \mathbb{R}^6 \quad \text{(NED frame)}$$

Control input: $\mathrm{u}_k = \mathrm{a}^n_{\mathrm{lin}}(t_k)$ – rotated & de-gravitized specific force from IMU.

Measurements:

$$\mathrm{z}_k = \begin{bmatrix} \mathrm{p}_n^{\mathrm{GNSS}} \\ \mathrm{v}_n^{\mathrm{GNSS}} \end{bmatrix}, \quad \mathrm{z}_k^{\mathrm{ZUPT}} = 0$$

Why this form? Simple, interpretable, and directly supported by GNSS. Attitude held fixed (Task 3).

Equations:

$$\mathsf{p}_{k+1} = \mathsf{p}_k + \mathsf{v}_k\,\Delta t$$
$$\mathsf{v}_{k+1} = \mathsf{v}_k + \mathsf{u}_k\,\Delta t$$

Compact form:

$$\mathsf{x}_{k+1} = \mathsf{F}\mathsf{x}_k + \mathsf{B}\mathsf{u}_k + \mathsf{w}_k \quad \text{where} \quad \mathsf{w}_k \sim \mathcal{N}(0, \mathsf{Q})$$

Matrices:

$$\mathsf{F} = \begin{bmatrix} \mathsf{I}_3 & \Delta t\,\mathsf{I}_3 \\ 0 & \mathsf{I}_3 \end{bmatrix}, \quad \mathsf{B} = \begin{bmatrix} \frac{1}{2}\Delta t^2\,\mathsf{I}_3 \\ \Delta t\,\mathsf{I}_3 \end{bmatrix}$$

Noise tuning: $q_{\text{pos}} = 0.002819, \quad q_{\text{vel}} = 1.1277$

Task 5.4: Measurement & Covariance Models

GNSS measurement:

$$\mathrm{z}_k = \mathrm{H}\mathrm{x}_k + \boldsymbol{\nu}_k, \quad \mathrm{H} = \begin{bmatrix} \mathrm{I}_3 & 0 \\ 0 & \mathrm{I}_3 \end{bmatrix}, \quad \boldsymbol{\nu}_k \sim \mathcal{N}(0, \mathrm{R})$$

ZUPT (during static):

$$\mathrm{z}_k^{\mathrm{ZUPT}} = \mathrm{H}_Z \mathrm{x}_k + \boldsymbol{\nu}_Z, \quad \mathrm{H}_Z = [0 \;\; \mathrm{I}_3]$$

From tuning:

$$\mathrm{R}_{\mathrm{GNSS}} = \mathrm{diag}(9.27,\; 9.68,\; 8.13,\; 25.0,\; 2.46,\; 25.0)$$

ZUPT activates when:

$$\|\tilde{\mathrm{a}}_b\| < 0.050, \quad \|\tilde{\boldsymbol{\omega}}_b\| < 0.005$$

Predict (IMU-rate):

1. $\tilde{a}_n(t) = R_{b\to n}\big(a_b(t) - b_a\big) - g_n$
2. $x_{k+1|k} = F x_{k|k} + B u_k$
3. $P_{k+1|k} = F P_{k|k} F^\top + Q$

Update (GNSS or ZUPT):

$$y_k = z_k - H x_{k|k-1}, \quad K_k = P_{k|k-1} H^\top \big(H P_{k|k-1} H^\top + R\big)^{-1}$$

$$x_{k|k} = x_{k|k-1} + K_k y_k, \quad P_{k|k} = (I - K_k H) P_{k|k-1}$$

Task 5.6: ZUPT Activation & Static Fraction

ZUPT trigger:

$$\|\tilde{a}_b\| < 0.050 \text{ and } \|\tilde{\boldsymbol{\omega}}_b\| < 0.005$$

From your run:

- ZUPT used: 479,621 times
- Static duration $\approx$ 95.9%

Effect:

- Sharp correction of drift
- Shrinks velocity covariance
- Stabilizes open-loop propagation

Task 5.7: Tuning Summary

Process noise (Q):

- $q_{\text{vel}} = 1.1277$ on velocity block
- $q_{\text{pos}} = 0.002819$ on position block

Measurement noise (R):

- Position: $\mathrm{diag}(9.27, 9.68, 8.13)$
- Velocity: $\mathrm{diag}(25.0, 2.46, 25.0)$

Other:

- Time alignment: 0.000 s
- Clipping on $v > 1500$ m/s (3510 events clipped)
- ZUPT count: 479,621

Task 5.8: Final Fused Result (Altitude Determination Method)

End kinematics (NED):

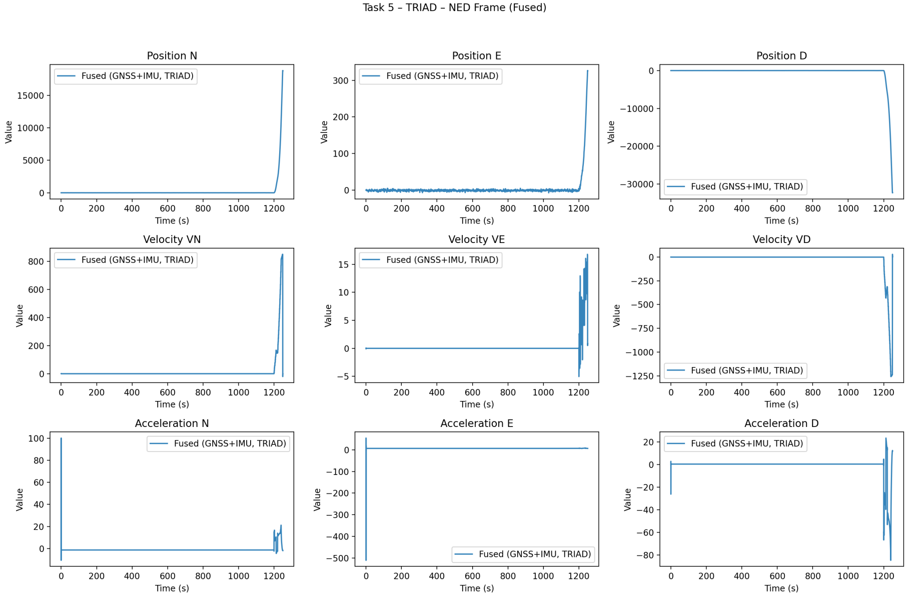
- $p_n = [18{,}786.9,\ 326.2,\ -32{,}273.1]$ m
- $v_n = [-0.0816,\ 0.8808,\ 1.3747]$ m/s
- $a_n = [-1.799,\ 6.757,\ 12.109]$ m/s^2

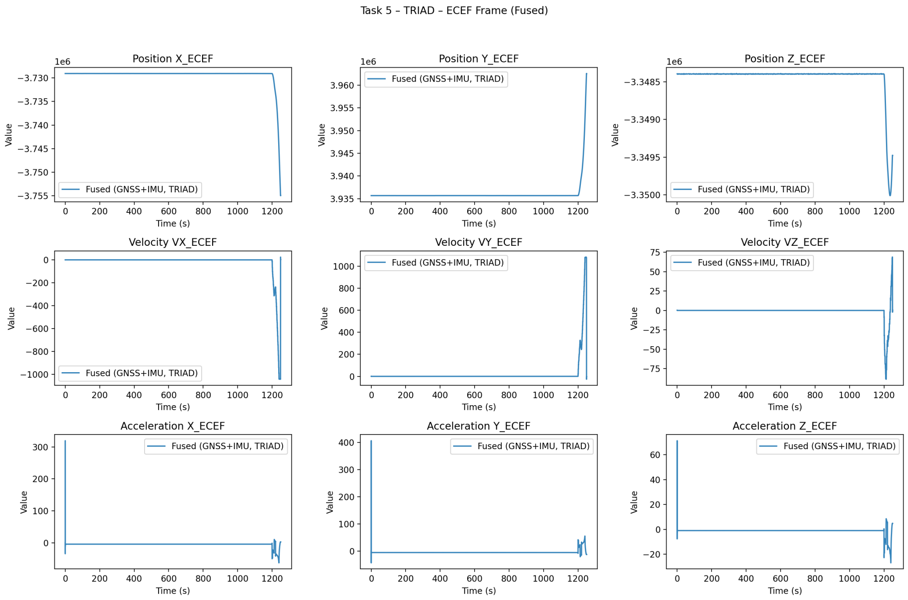
Errors:

- RMSE (position): 0.94 m
- Final position error: 0.08 m
- RMS velocity residual: 5.61 m/s (max $=$ 123.94 m/s)

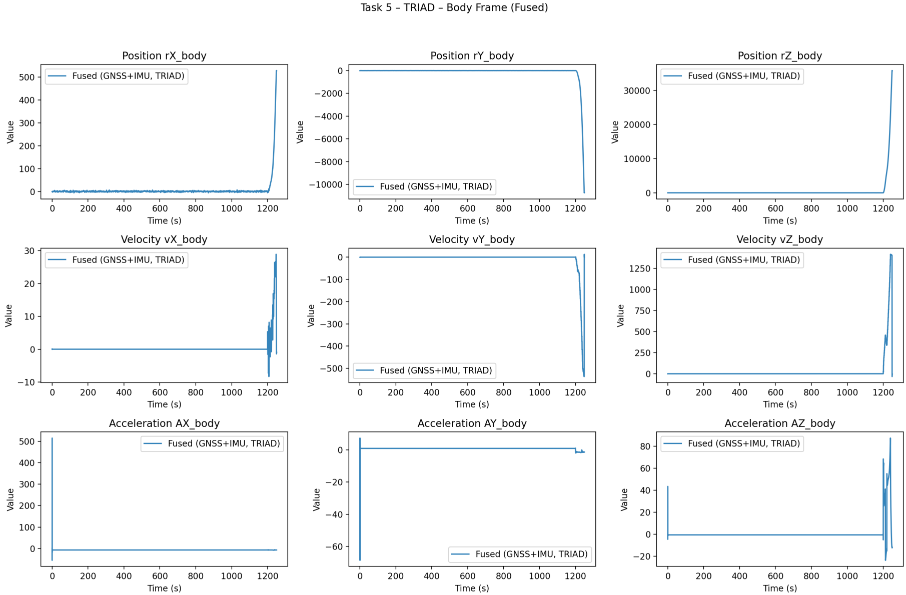
Usage:

- GNSS used: 480,067 times (96%)
- RMS GNSS $|v|$: 187.62 m/s


Task 5 – TRIAD – NED Frame (Fused)
Position N
Position E
Position D
Velocity VN
Velocity VE
Velocity VD
Acceleration N
Acceleration E
Acceleration D
Fused (GNSS+IMU, TRIAD)
Value
Time (s)


Task 5 – TRIAD – ECEF Frame (Fused)
Position X_ECEF
1e6
−3.730
−3.735
−3.740
−3.745
−3.750
−3.755
Value
Fused (GNSS+IMU, TRIAD)
Position Y_ECEF
1e6
3.960
3.955
3.950
3.945
3.940
3.935
Position Z_ECEF
1e6
−3.3485
−3.3490
−3.3495
−3.3500
Velocity VX_ECEF
Velocity VY_ECEF
Velocity VZ_ECEF
Acceleration X_ECEF
Acceleration Y_ECEF
Acceleration Z_ECEF
Time (s)
0
200
400
600
800
1000
1200

Task 5.10: Body Frame – IMU Behavior



Task 5.11: Interpretation & Limitations

What works well:

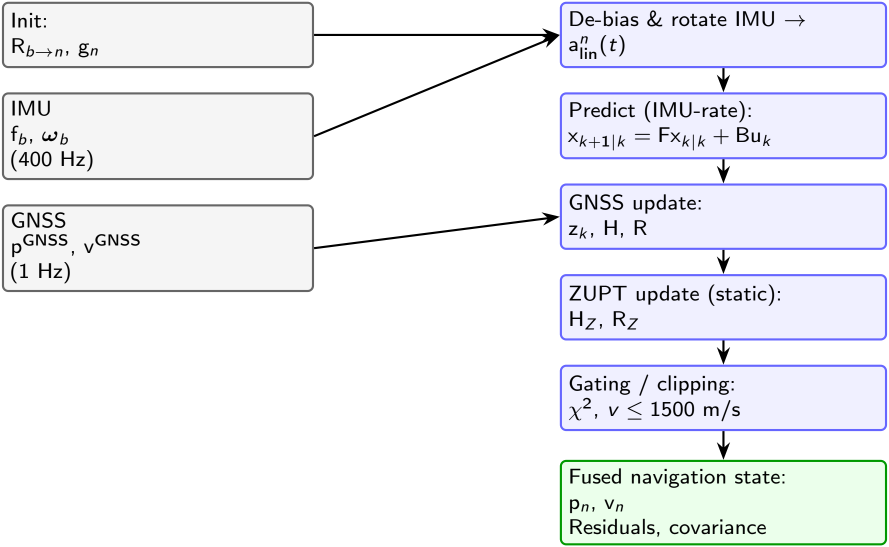
- GNSS + ZUPT keeps drift bounded
- Static phase gives reliable bias correction
- Attitude frame held from Task 3, stable

Limitations:

- GNSS velocity is noisy (1 Hz finite difference)
- Static data $\Rightarrow$ limited dynamic observability

Next: Validate with the TRUTH trajectory

Task 5.12: Box Diagram — Filter Pipeline



Legend: Gray = inputs Blue = processing Green = outputs

Task 7: Body Frame – Fused vs. Truth

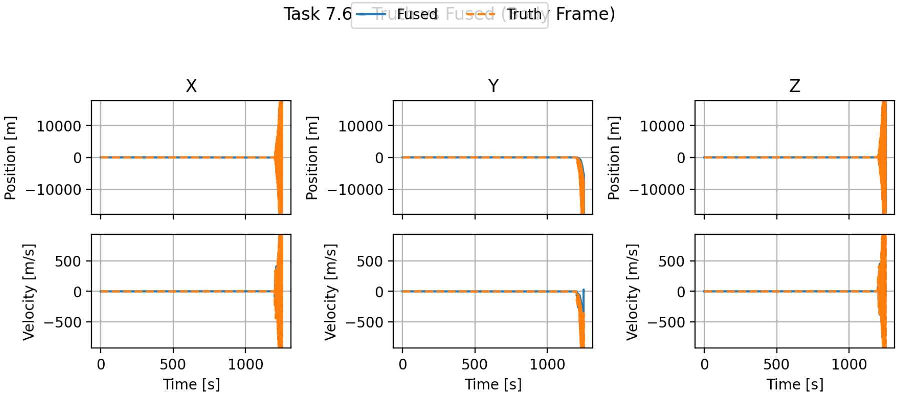


Figure: Body-frame Truth vs. Fused

Task 7: ECEF Frame – Fused vs. Truth

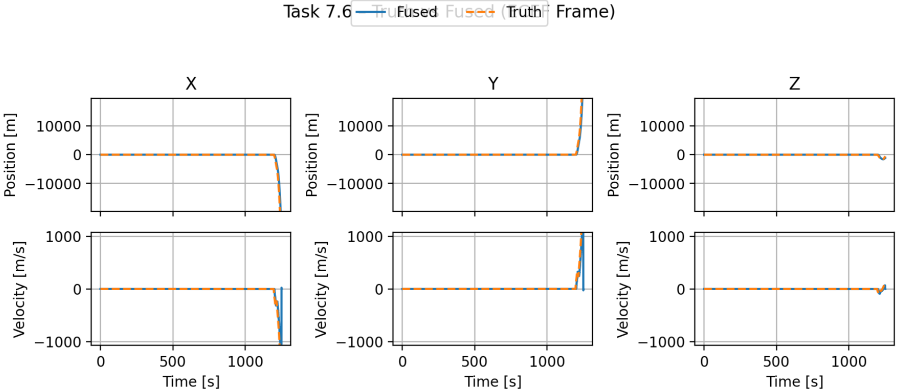


Figure: ECEF-frame Truth vs. Fused

Task 7: NED Frame – Fused vs. Truth

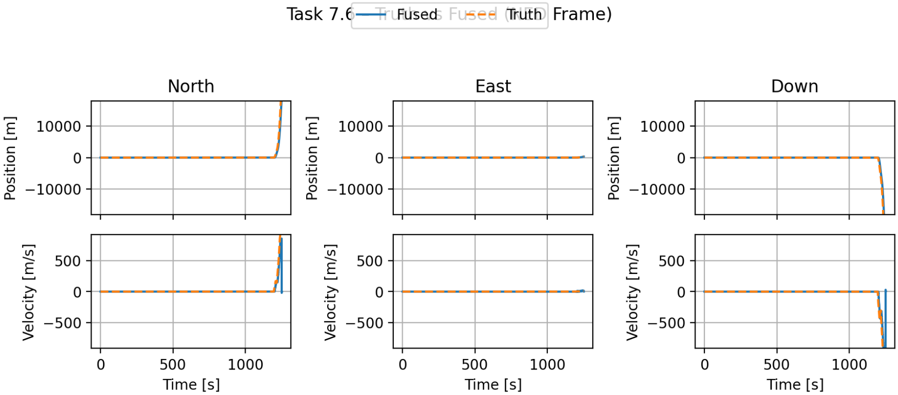
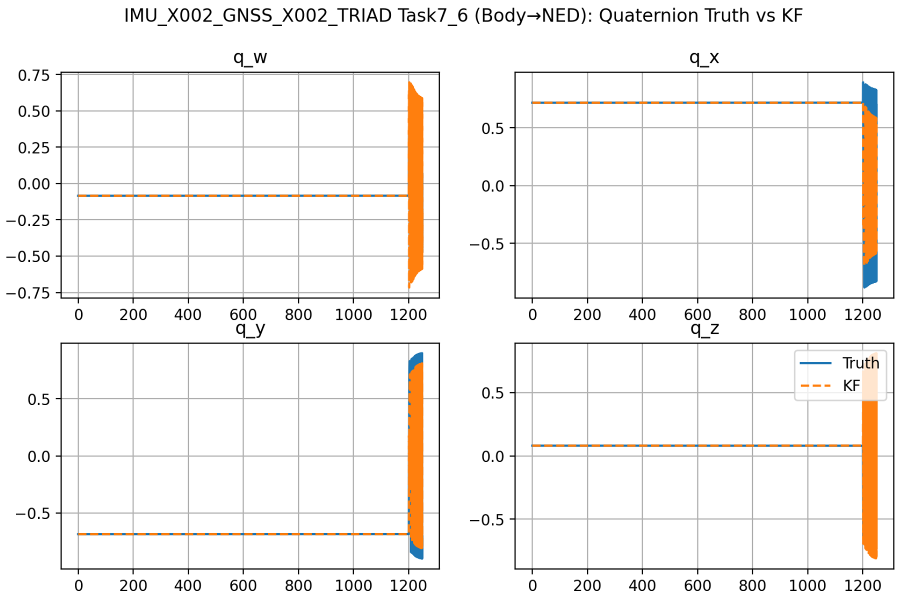


Figure: NED-frame Truth vs. Fused

Task 7: Quaternions Frame – Fused vs. Truth



Task 7: Difference

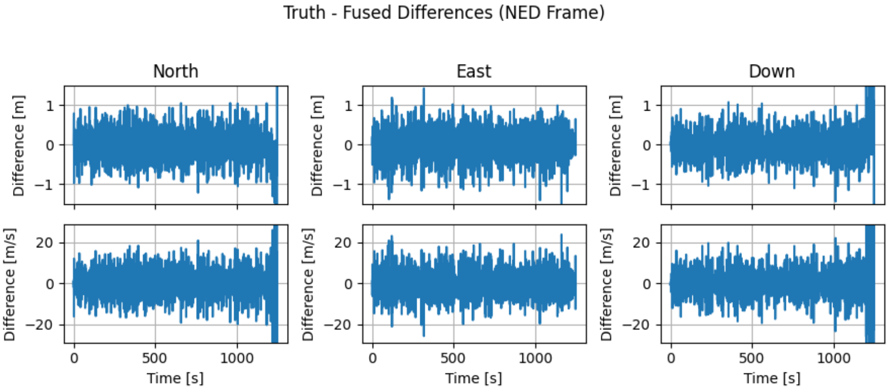


Figure: Difference

Task 7: Difference

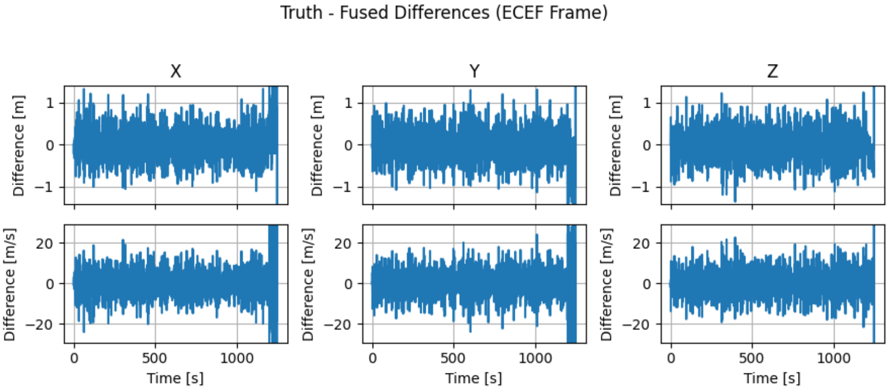


Figure: Difference

Task 7: Difference

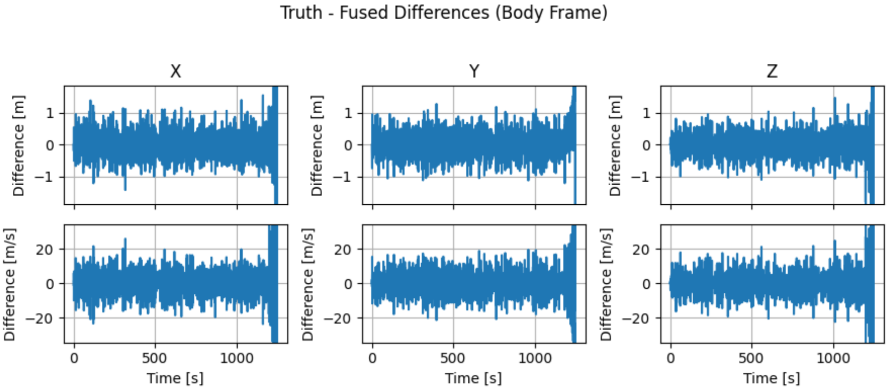


Figure: Difference

Thank you !